

Exp no:8 -Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

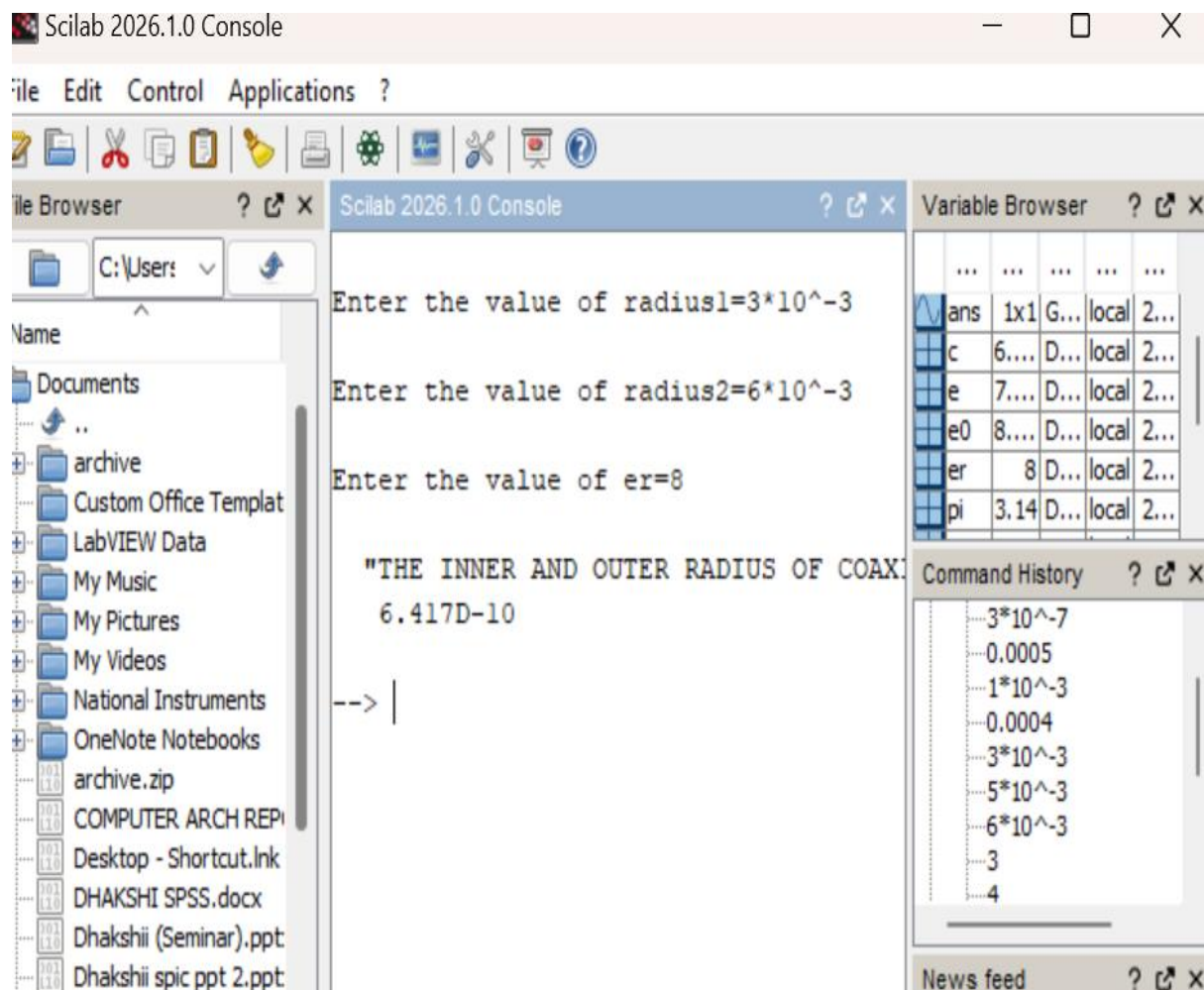
Aim:

To write program for Capacitance of co-axial cable per unit length using SCILAB.

Software required:

- SCILAB version 6.0.1

Partical calucation:



exp 8.sci (C:\Users\dhakshinya velan\Documents\exp 8.sci) - SciNotes

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exp 8.sci (C:\Users\dhakshinya velan\Documents\exp 8.sci) - SciNotes

EMF EXPRIMENT 2.sci X EXP6.sci X exp 7.sci X exp 8.sci X

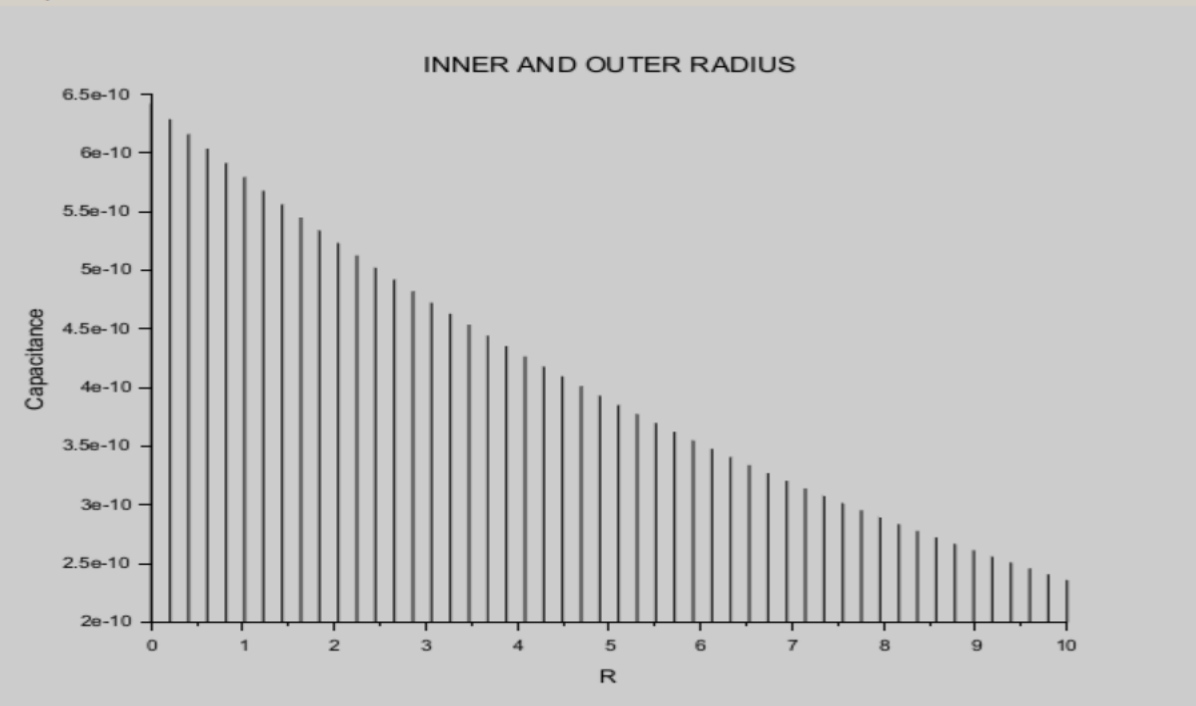
```
1 clc;
2 clear;
3 r1=input("Enter the value of radius1=");
4 r2=input("Enter the value of radius2=");
5 er=input("Enter the value of er=");
6
7 r=r2/r1;
8 e0=8.854*10^-12;
9 pi=3.14;
10 e=e0*er;
11
12 c=(2*pi*e)/(log(r));
13 disp('THE INNER AND OUTER RADIUS OF COAXIAL CABLE=',[c]);
14
15 x=linspace(0,10,50);
16 y=c*exp(-0.1*x);
17
18 figure();
19 subplot(1,1,1);
20 plot2d3(x,y);
21 xlabel("R");
22 ylabel("Capacitance");
```

Graphic window number 1

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Graphic window number 1



Theoretical calculation:

Handwritten calculations for the capacitance of a co-axial cable. The calculations are as follows:

$$\begin{aligned} 8) \quad r_1 &= 3 \text{ mm} \\ r_2 &= 6 \text{ mm} \\ \epsilon_r &= 8 \\ \epsilon_0 &= 8.85 \times 10^{-12} \\ C &= (8.85 \times 10^{-12}) / (8) = 7.0832 \times 10^{-11} \\ C &= 2\pi\epsilon / \log(r) = \log(2) = 0.6931 \\ C &= \frac{2(3.14)(7.0832 \times 10^{-11})}{0.6931} \\ &= \frac{2(3.14)(7.0832 \times 10^{-11})}{0.6931} = 4.448 \times 10^{-10} \\ C &= 6.417 \times 10^{-10} \end{aligned}$$

Result:

Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.